US012398822B2

(12) **United States Patent**
Ejima et al.

(10) **Patent No.: US 12,398,822 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **VALVE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/267,072**

(22) PCT Filed: **Dec. 6, 2021**

(86) PCT No.: **PCT/JP2021/044726**

§ 371 (c)(1),
(2) Date: **Jun. 13, 2023**

(87) PCT Pub. No.: **WO2022/131053**

PCT Pub. Date: **Jun. 23, 2022**

(65) **Prior Publication Data**

US 2024/0011574 A1 Jan. 11, 2024

(30) **Foreign Application Priority Data**

Dec. 17, 2020 (JP) 2020-209652

(51) **Int. Cl.**
F16K 31/06 (2006.01)
F16K 1/46 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***F16K 31/0655*** (2013.01); ***F16K 1/46*** (2013.01); *B60H 1/00485* (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC F16K 1/46; F16K 31/0655; F16K 1/34; F16K 31/0613; F16K 5/207;
(Continued)

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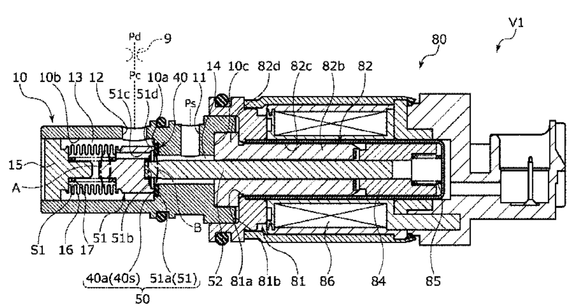
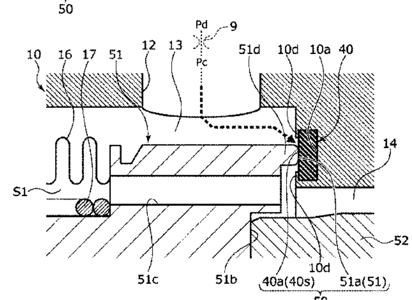
(57) **ABSTRACT**

There is provided a valve capable of reducing valve leakage. The valve includes: a valve housing in which ports through which a fluid passes are formed; a valve body configured to be driven by a drive source; a valve seat on which a contact portion of the valve body is seated; and biasing members for biasing the valve body in a valve closing direction. The valve seat is formed of an elastic member.

15 Claims, 9 Drawing Sheets





(51) **Int. Cl.**
B60H 1/00 (2006.01)
F04B 27/18 (2006.01)
F25B 41/345 (2021.01)

(52) **U.S. Cl.**
CPC *F04B 27/1804* (2013.01); *F16K 31/0613* (2013.01); *F25B 41/345* (2021.01)

(58) **Field of Classification Search**
CPC F16K 17/087; F16K 31/126; F04B 2027/1818; F04B 2027/1831; F04B 2027/1859; F04B 27/1804; F04B 2027/1822–1881; B60H 1/00485; F25B 41/345
USPC 137/625.65; 417/222.2; 251/359–363, 251/129.15
See application file for complete search history.

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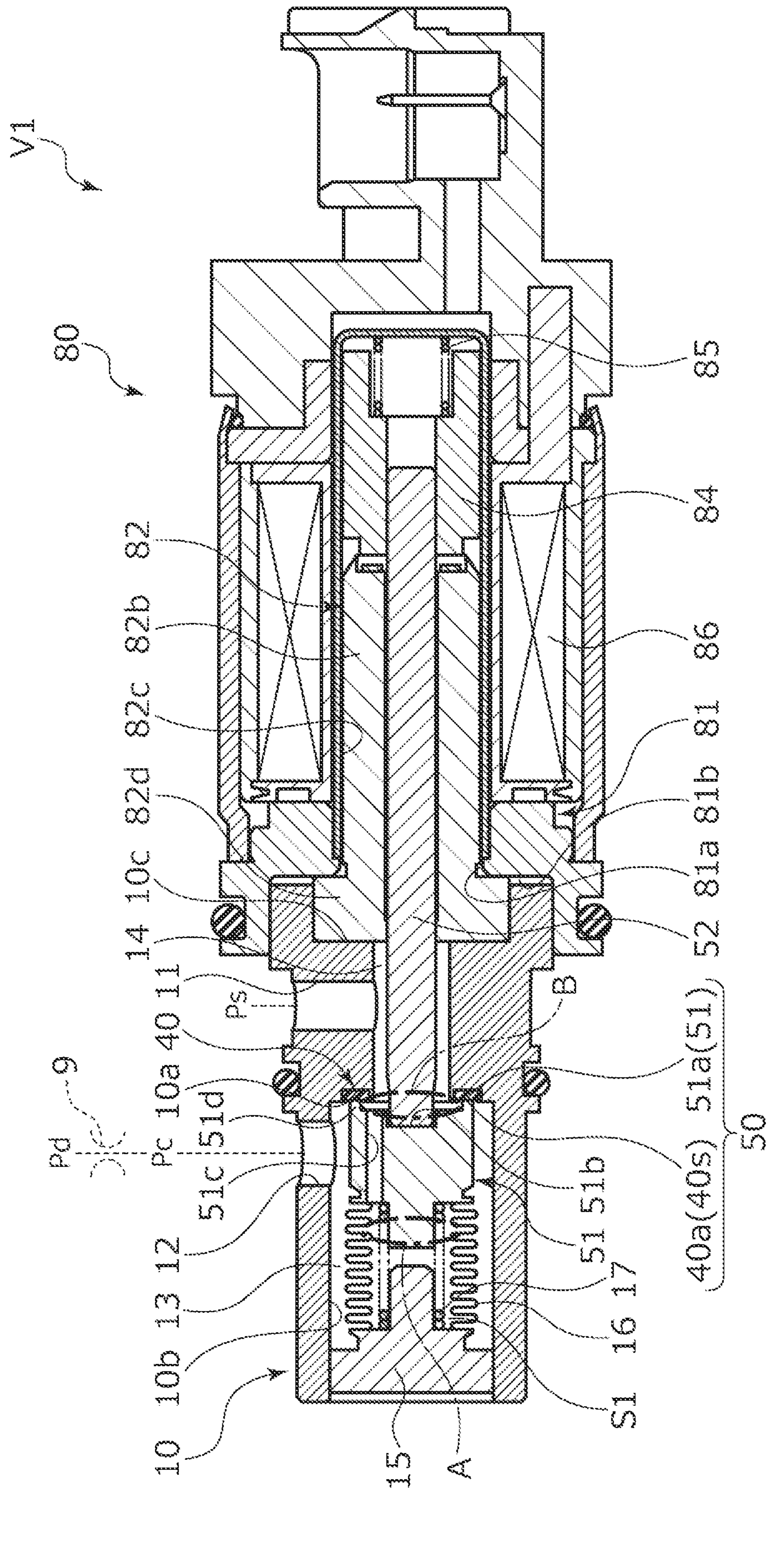
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84
86
81
81b
81a
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51b
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17
16
S1
A
15
10
10b
13
12
51c
Pc
Pd
9
51d
10a
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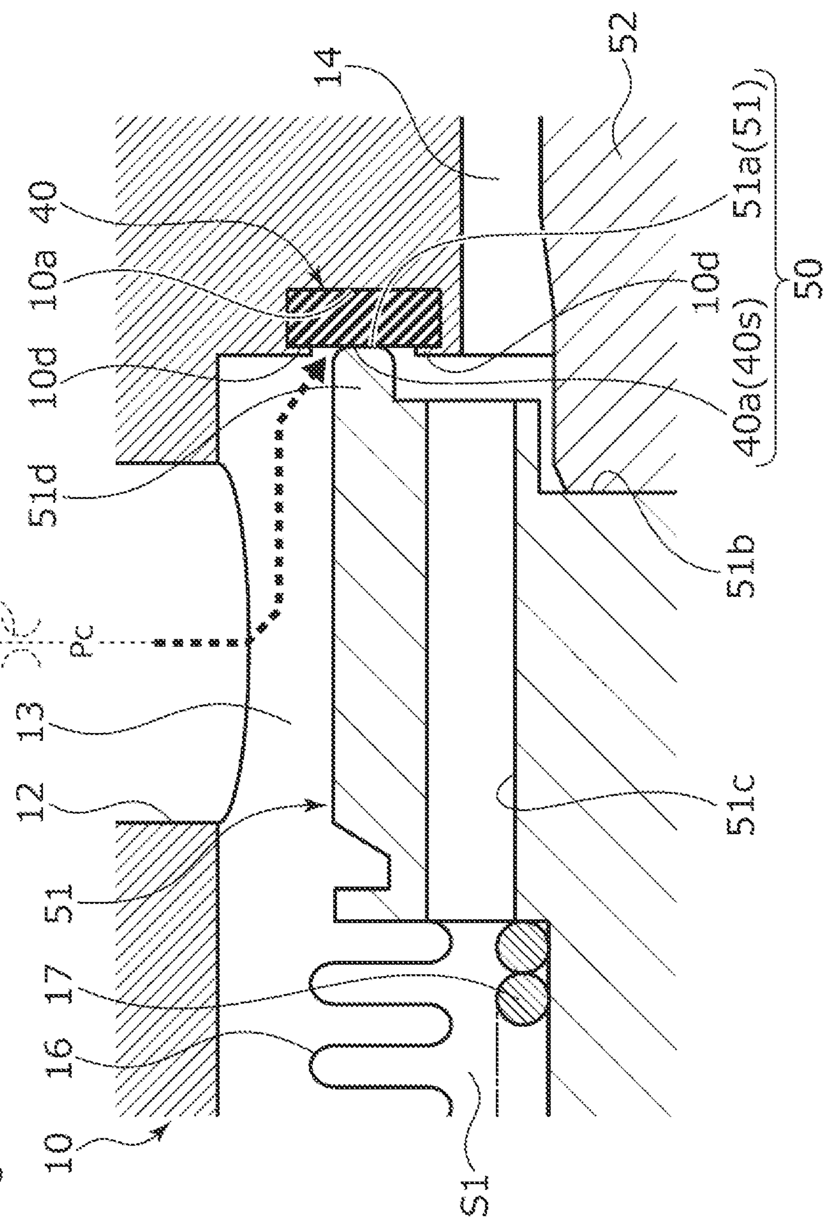
82b
82

Fig. 1

Fig.2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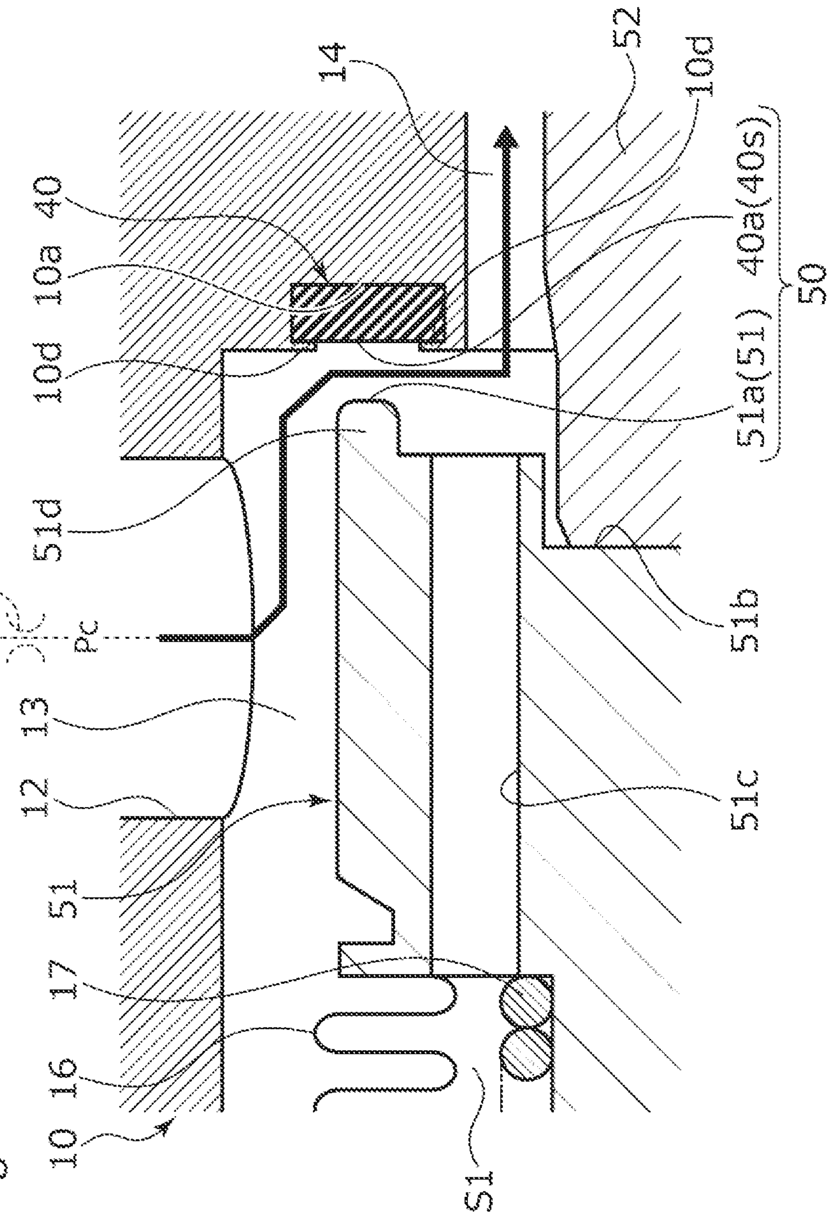
16
17
51
12
13
Pc
Pd
9
51d
10d
10a
40
14
52
51a(51)
10d
40a(40s)
50
51b
51c
S1

Fig.3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17
51
12
13
Pd
Pc
9
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10d
10a
40
14
52
10d
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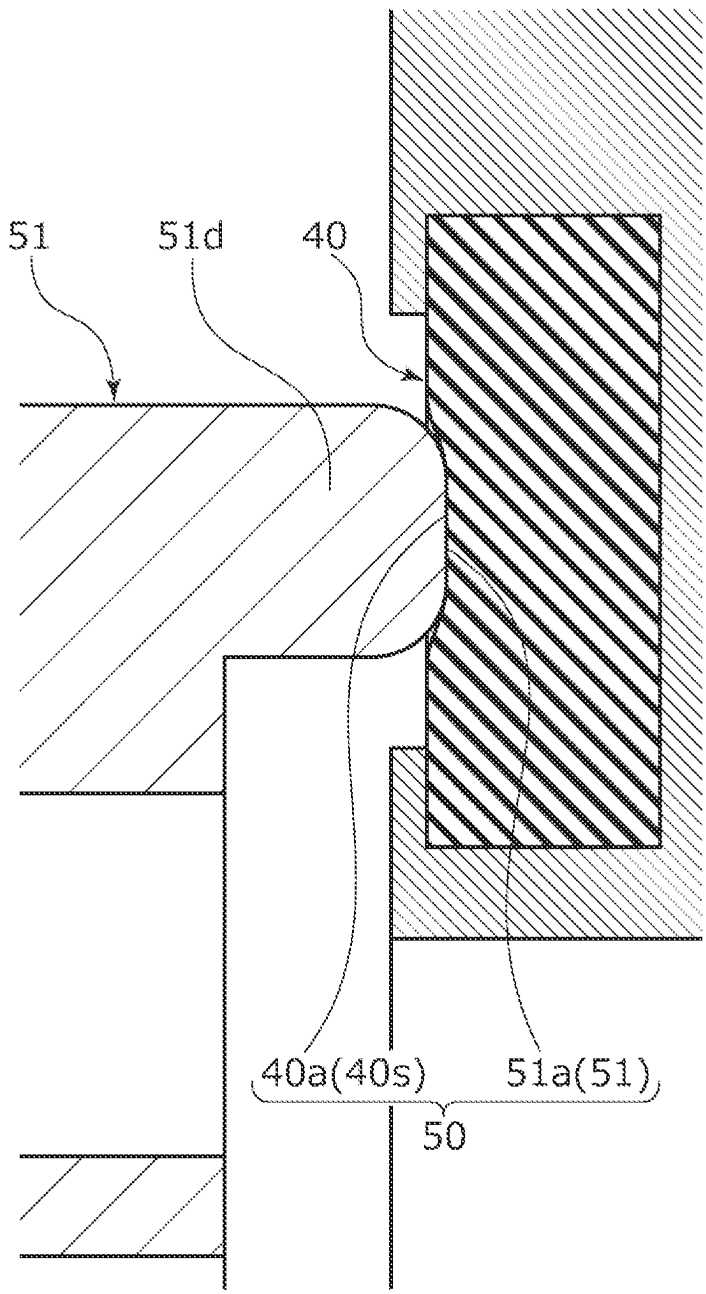
51a(51)
50
51b
51c
S1

Fig.4



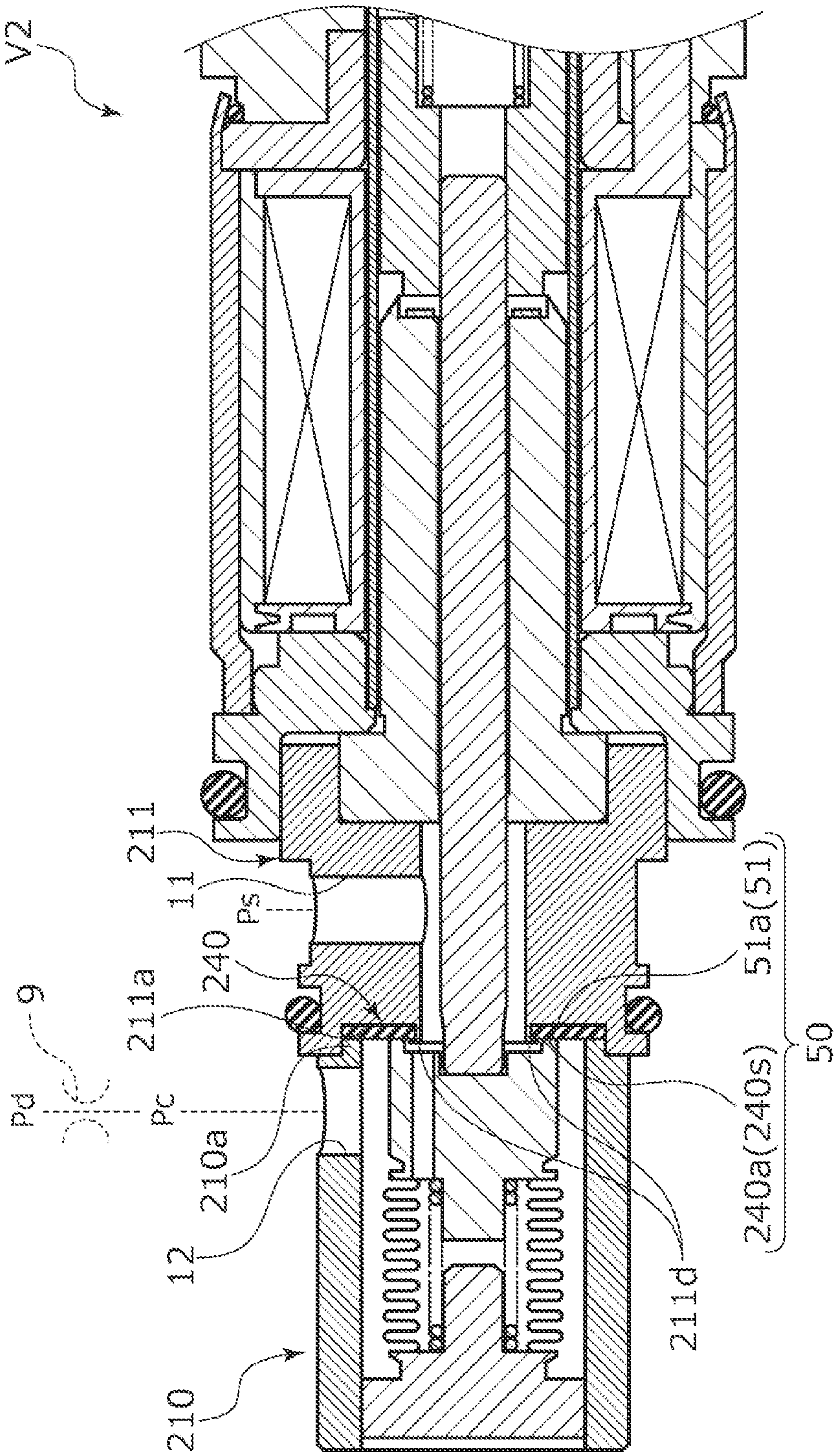
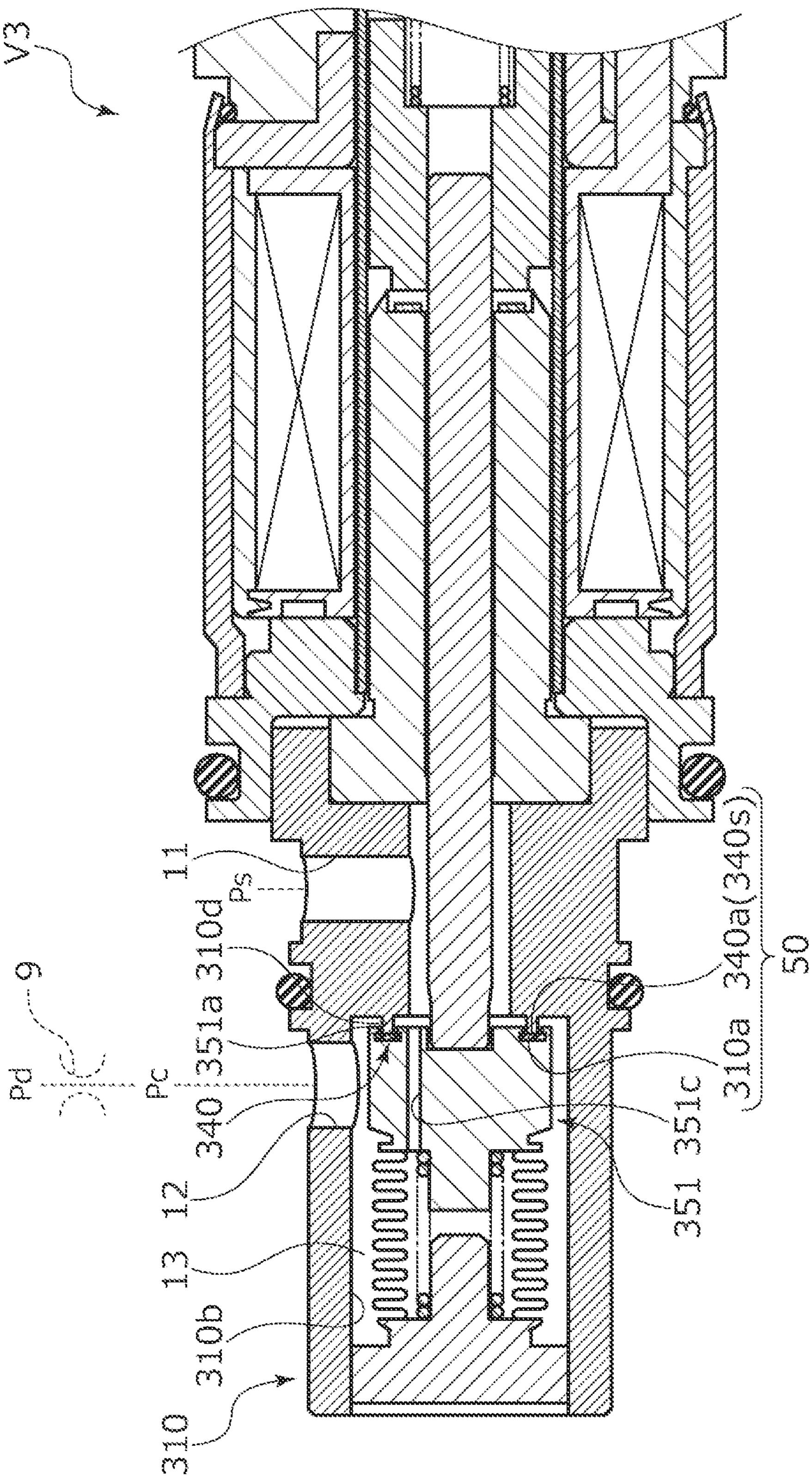


Fig. 5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Pd
Pc
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Ps
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340
12
13
310b
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340a(340s)
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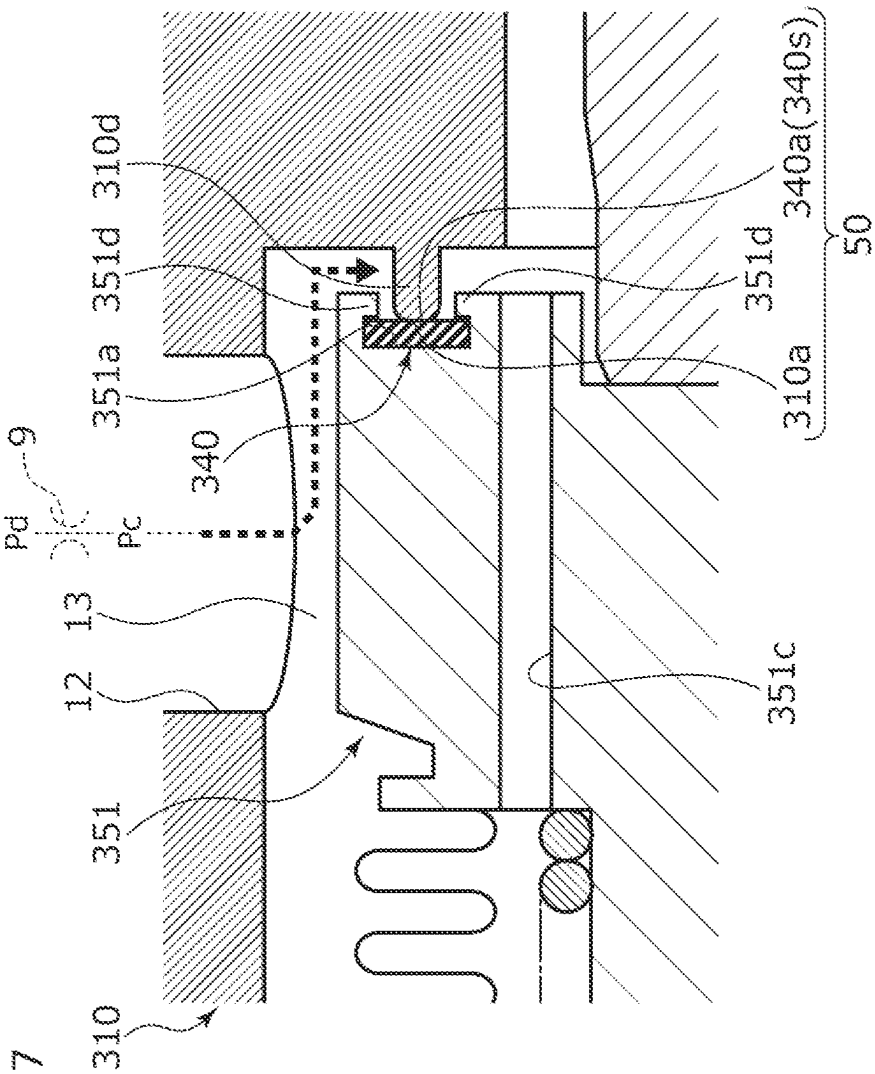
351c
351

Fig. 6

Fig. 7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Pc
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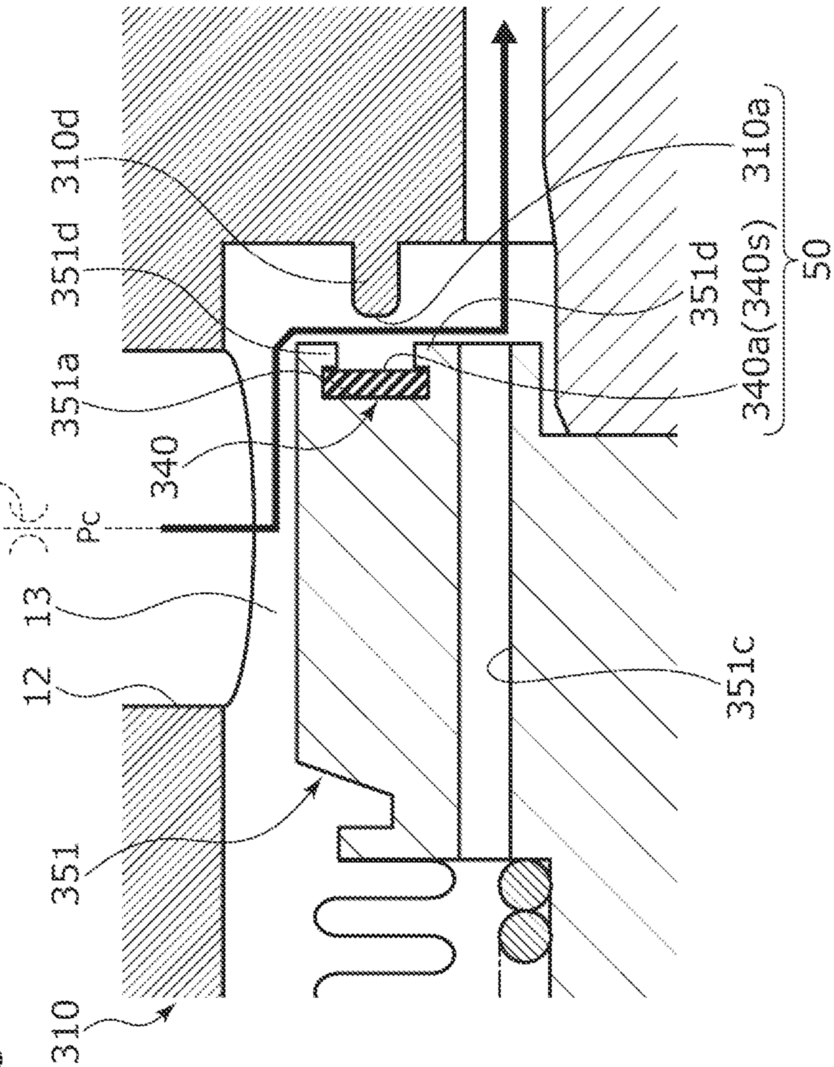
351d
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310a
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351a
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Pd
Pc
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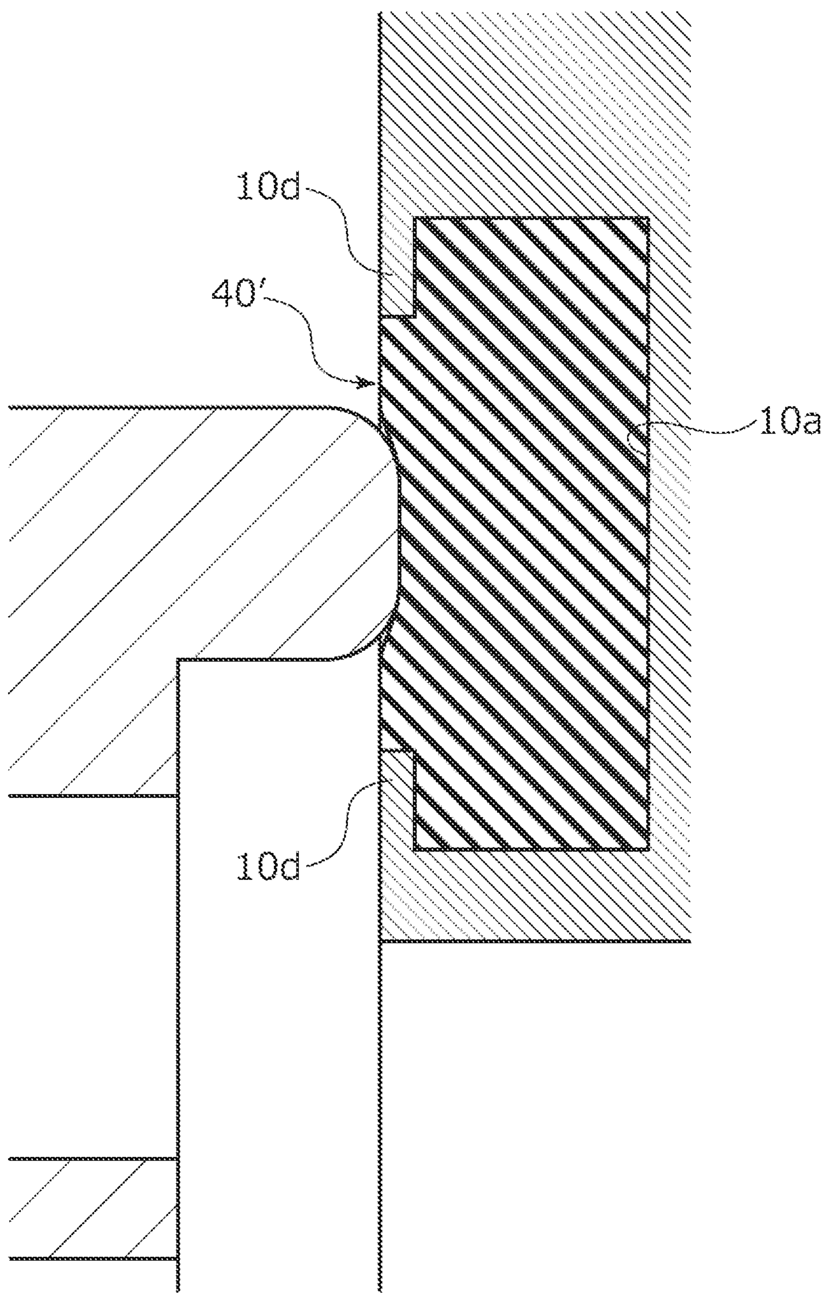
351c
351
310

Fig.8

Fig.9



VALVE

TECHNICAL FIELD

The present invention relates to a valve that variably controls a working fluid, for example, to a valve that controls a discharge amount of a variable capacity compressor used for an air conditioning system of an automobile according to pressure.

BACKGROUND ART

A variable capacity compressor used for an air conditioning system of an automobile, etc. includes a rotating shaft to be rotationally driven by an engine, a swash plate coupled to the rotating shaft in such a manner that a tilt angle is variable, compressing pistons coupled to the swash plate, etc., and by changing the tilt angle of the swash plate, changes a stroke amount of the pistons to control a discharge amount of fluid. This tilt angle of the swash plate can be continuously changed by appropriately controlling pressure in a control chamber while utilizing a suction pressure Ps of a suction chamber that suctions the fluid, a discharge pressure Pd of a discharge chamber that discharges the fluid pressurized by the pistons, and a control pressure Pc of the control chamber that houses the swash plate, by means of a capacity control valve as a valve that is driven to open and close by electromagnetic force of a solenoid as a drive source.

At the time of continuously driving the variable capacity compressor, the capacity control valve performs normal control in which energization is controlled by a control computer, a valve body is moved in an axial direction by electromagnetic force generated in the solenoid, and a flow passage between a discharge port and a control port is opened and closed by a valve to adjust the control pressure Pc of the control chamber of the variable capacity compressor.

In addition, there is a capacity control valve that controls a flow rate of the fluid flowing from the control port to a suction port. For example, in a capacity control valve disclosed in Patent Citation 1, in an open state of the valve in which the solenoid is energized, the fluid flows to the suction port through a through-flow passage communicating with the control port inside a housing. When the solenoid is de-energized from the open state, a valve body having a rod shape can be moved toward a valve seat by biasing force of a bellows to close the through-flow passage, the valve seat being formed in the valve housing. In such a manner, pressure in the control chamber of the variable capacity compressor is controlled using a pressure difference between the control pressure Pc and the suction pressure Ps lower than the control pressure Pc.

CITATION LIST

Patent Literature

Patent Citation 1: WO 2020/218284 A (Pages 11 to 13, FIG. 4)

SUMMARY OF INVENTION

Technical Problem

In the capacity control valve of Patent Citation 1, a contact portion having a tapered shape is formed such that

one end of the valve body is tapered, and comes into contact with and is seated on the valve seat formed on an inner peripheral surface of the valve housing, through a wide surface. However, operation in which the valve body returns toward the valve seat from an open state is due to biasing force of the bellows, and return force of the valve body that depends on biasing force of the bellows is small. For this reason, in a case where contaminations are caught between the contact portion of the valve body and the valve seat when the valve is closed, the contaminations are not completely crushed, and a gap occurs between the contact portion of the valve body and the valve seat and becomes a cause of valve leakage, which is a problem.

The present invention is conceived in view of such a problem, and an object of the present invention is to provide a valve capable of reducing valve leakage.

Solution to Problem

In order to solve the foregoing problem, a valve according to the present invention includes: a valve housing in which a port through which a fluid passes is formed; a valve body to be driven by a drive source; a valve seat on which a contact portion of the valve body is seated; and a biasing member configured to bias the valve body in a valve closing direction or the valve. At least one of the contact portion of the valve body and the valve seat is formed of an elastic member. According to the aforesaid feature of the present invention, when the valve is closed, even in a case where contaminations are caught between the contact portion of the valve body and the valve seat, the elastic member can be elastically deformed to suppress generation of a gap between the contact portion and the valve seat, so that valve leakage can be reduced.

It may be preferable that the contact portion of the valve body or the valve seat that comes into contact with a contact surface of the elastic member is formed of an annular projection and the elastic member is formed to have a modulus of elasticity smaller than a modulus of elasticity of the annular projection. According to this preferable configuration, when the valve is closed, even in a case where contaminations are caught between the annular projection and the contact surface of the elastic member, the elastic member can be reliably and elastically deformed to suppress generation of a gap between the contact portion and the valve seat, so that valve leakage can be reduced.

It may be preferable that the contact surface is a surface orthogonal to a driving direction of the valve body. According to this preferable configuration, when the valve is closed, since it is difficult for the annular projection to relatively move with respect to the contact surface of the elastic member, sealability is enhanced, and damage to the contact surface of the elastic member caused by the annular projection is suppressed, so that sealability can be maintained over a long period of time.

It may be preferable that the valve body or the valve housing is provided with an annular recessed portion and the elastic member inserted into the annular recessed portion is crimped and fixed from at least one of radially inner and outer sides. According to this preferable configuration, the elastic member inserted into the annular recessed portion can be prevented from coming off.

It may be preferable that the elastic member has a rectangular cross section. According to this preferable configuration, even when the annular projection comes into contact with the contact surface of the elastic member at any position, stable elasticity can be provided.

It may be preferable that the valve body is separately configured to come into contact with and to separate from a rod forming the drive source and the rod is biased in a valve opening direction by a rod biasing member. According to this preferable configuration, since the rod is held so as to be able to come into contact with and to separate from the valve body in a state where the rod is biased in the valve opening direction, when the valve is closed, the valve body is not affected by an inertial force of the rod, so that an excessive load can be prevented from being applied to the contact portion or to the valve seat.

It may be preferable that the biasing member is a compression spring. According to this preferable configuration, a structure on a drive source side of the valve is simplified, and the contact portion of the valve body is seated on the valve seat while the structure is such that an axis of the valve body is easily offset, so that good sealability can be obtained.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a cross-sectional view illustrating a structure of a capacity control valve as a valve according to a first embodiment of the present invention.

FIG. **2** is an enlarged cross-sectional view illustrating a state where a CS valve is closed in the capacity control valve in the first embodiment.

FIG. **3** is an enlarged cross-sectional view illustrating a state where the CS valve is opened in the capacity control valve in the first embodiment.

FIG. **4** is an enlarged cross-sectional view illustrating a state where an elastic member is elastically deformed when the CS valve is closed in the first embodiment.

FIG. **5** is an enlarged cross-sectional view illustrating a structure of a capacity control valve as a valve according to a second embodiment of the present invention.

FIG. **6** is an enlarged cross-sectional view illustrating a structure of a capacity control valve as a valve according to a third embodiment of the present invention.

FIG. **7** is an enlarged cross-sectional view illustrating a state where a CS valve is closed in the capacity control valve in the third embodiment.

FIG. **8** is an enlarged cross-sectional view illustrating a state where the CS valve is opened in the capacity control valve inf the third embodiment.

FIG. **9** is a view illustrating a modification example of the elastic member.

DESCRIPTION OF EMBODIMENTS

Modes for implementing a valve according to the present invention will be described below based on embodiments. Incidentally, in the embodiments, a capacity control valve will be described as an example but the present invention is also applicable to other uses such as an expansion valve.

First Embodiment

A capacity control valve as a valve according to a first embodiment of the present invention will be described with reference to FIGS. **1** to **3**. Hereinafter, left and right sides when viewed from the front side of FIG. **1** will be described as being left and right sides of the capacity control valve. Specifically, the left side of the drawing sheets on which a valve housing **10** is disposed and the right side of the drawing sheets on which a solenoid **80** is disposed will be described as being the left side of the capacity control valve and the right side of the capacity control valve, respectively.

The capacity control valve of the present invention is assembled into a variable capacity compressor (not illustrated) used for an air conditioning system of an automobile, etc., and variably controls pressure of a working fluid (hereinafter, simply referred to as a "fluid") that is a refrigerant. Accordingly, the capacity control valve controls a discharge amount of the variable capacity compressor to adjust the air conditioning system to a target cooling capacity.

First, the variable capacity compressor will be described. The variable capacity compressor includes a casing including a discharge chamber, a suction chamber, a control chamber, and a plurality of cylinders. Incidentally, the variable capacity compressor is provided with a communication passage that provides direct communication between the discharge chamber and the control chamber. The communication passage is provided with a fixed orifice **9** that balances the pressures of the discharge chamber and the control chamber (refer to FIG. **1**).

In addition, the variable capacity compressor includes a rotating shaft, a swash plate, and a plurality of pistons. The rotating shaft is rotationally driven by an engine (not illustrated) installed outside the casing. The swash plate is coupled to the rotating shaft so as to be tiltable with respect to the rotating shaft by a hinge mechanism in the control chamber. The plurality of pistons are coupled to the swash plate and are reciprocatably fitted in the respective cylinders. A capacity control valve V**1** is driven to open and close by electromagnetic force, so that pressure in the control chamber of a variable capacity compressor is appropriately controlled using a suction pressure Ps of the suction chamber that suctions the fluid, a discharge pressure Pd of the discharge chamber that discharges the fluid pressurized by the pistons, and a control pressure Pc of the control chamber that houses the swash plate. Accordingly, a tilt angle of the swash plate changes continuously. Accordingly, a stroke amount of the pistons is changed, so that a discharge amount of the fluid from the variable capacity compressor is controlled.

As illustrated in FIG. **1**, the capacity control valve V**1** of the first embodiment assembled into the variable capacity compressor adjusts an electric current that energizes a coil **86** forming the solenoid **80** as a drive source, to perform opening and closing control of a CS valve **50** in the capacity control valve V**1**. Accordingly, the fluid flowing out from the control chamber to the suction chamber is adjusted to variably control the control pressure Pc in the control chamber. Incidentally, a discharge fluid of the discharge pressure Pd of the discharge chamber is constantly supplied to the control chamber via the fixed orifice **9**, and the CS valve **50** in the capacity control valve V**1** is closed, so that the control pressure Pc in the control chamber is increased.

In the capacity control valve V**1** of the first embodiment, the CS valve **50** is formed of a CS valve body **51** as a valve body and a CS valve seat **40***a* as a valve seat. The CS valve seat **40***a* is formed on an elastic member **40** that is press-fitted and crimped and fixed to an annular recessed portion **10***a* of the valve housing **10**. A contact portion **51***a* formed at an axially right end of the CS valve body **51** comes into contact with and separates from the CS valve seat **40***a* in an axial direction to open and close the CS valve **50**.

Next, a structure of the capacity control valve V**1** will be described. As illustrated in FIG. **1**, the capacity control valve V**1** mainly includes the valve housing **10** made of a metallic material; the CS valve body **51** disposed to be reciprocatable in the axial direction inside the valve housing **10**; and the solenoid **80** connected to the valve housing **10** to exert a driving force on the CS valve body **51**.

As illustrated in FIGS. **1** to **3**, the CS valve body **51** is made of a metallic material or a resin material, and includes a recessed portion **51***b* at a central portion thereof, the recessed portion **51***b* being open to the right in the axial direction. A rod **52** is disposed through the coil **86** of the solenoid **80**. An axially left end portion of the rod **52** is inserted into the recessed portion **51***b* so as to be contactable and separable. In addition, a communication passage **51***c* penetrating through the CS valve body **51** in the axial direction is formed in the CS valve body **51** at a position that is further offset in a radially outward direction than the recessed portion **51***b*. The communication passage **51***c* is formed with a constant cross section. Incidentally, a plurality of the communication passages **51***c* may be provided.

In addition, an annular projection **51***d* protruding to the right in the axial direction is formed on the CS valve body **51** at a position that is further offset in the radially outward direction than the communication passage **51***c*. A tip, namely, an axially right end of the annular projection **51***d* servers as the contact portion **51***a* that comes into contact with and separates from the CS valve seat **40***a* in the axial direction. In addition, the contact portion **51***a* that is the tip of the annular projection **51***d* is formed with a cross-sectional shape in which radially inner and outer sides of the contact portion **51***a* are round chamfered and a flat portion is provided between both round chamfers (refer to FIGS. **2** and **3**). Incidentally, the contact portion **51***a* may be formed with a substantially curved cross-sectional shape in which a flat portion is not provided between both round chamfers, or may be C-chamfered instead of being round chamfered. In addition, chamfering is not essential, and a chamfer may be formed only on one of the radially inner and outer sides or may be not formed.

As illustrated in FIGS. **1** to **3**, in the valve housing **10**, a Ps port **11** as a port penetrating through the valve housing **10** in a radial direction and communicating with the suction chamber of the variable capacity compressor, and a Pc port **12** as a port communicating with the control chamber are formed. The Ps port **11** is formed on an axially right side of the CS valve seat **40***a*, namely, in a valve closing direction to be described later. In addition, the Pc port **12** is formed on an axially left side of the CS valve seat **40***a*, namely, in a valve opening direction to be described later.

A first valve chamber **13** to which the fluid is supplied from the Pc port **12**, and a second valve chamber **14** to which the fluid that has passed through the CS valve **50** from the first valve chamber **13** is supplied and which communicates with the Ps port **11** are provided inside the valve housing **10**. The first valve chamber **13** is formed by a recessed portion **10***b* that is formed on the axially left side of the CS valve seat **40***a* and that is open to the left in the axial direction, and an opening portion on an axially left side is closed in a sealed manner by a lid member **15**.

In addition, a bellows **16** as a biasing member (also referred to as biasing means) for biasing the CS valve body **51** to the right in the axial direction, namely, in the valve closing direction, and a coil spring **17** as another biasing member are disposed in the first valve chamber **13**. An axially left end of the bellows **16** is fixed to the lid member **15** in a sealed manner, an axially right end of the bellows **16** is fixed to an axially left end surface of the CS valve body **51** in a sealed manner, and a space S1 is formed inside the bellows **16**. Incidentally, the coil spring **17** is a compression spring, and is disposed in the space S1 formed inside the bellows **16**.

In addition, the space S1 communicates with the second valve chamber **14** through the communication passage **51***c*, and the fluid in the second valve chamber **14** flows into the space S1. Namely, in a closed state of the CS valve **50**, the bellows **16** partitions the space S1 and the first valve chamber **13** off from each other in a sealed manner.

In addition, a recessed portion **10***c* of which a radially inner side of an axially right end is recessed to the left in the axial direction is formed in the valve housing **10**. A flange portion **82***d* of a center post **82** is inserted into the recessed portion **10***c* from the right in the axial direction, so that the center post **82** is connected and fixed to the valve housing **10** in a substantially sealed state. Incidentally, an opening end on a solenoid **80** side of the second valve chamber **14** is formed on a radially inner side of a bottom surface of the recessed portion **10***c* of the valve housing **10**.

Here, the elastic member **40** will be described. As illustrated in FIGS. **1** to **3**, the elastic member **40** is a ring having a rectangular cross section that is made of a material such a rubber or resin having a modulus of elasticity smaller than that of the annular projection **51***d* of the CS valve body **51**.

In addition, the elastic member **40** is press-fitted to the annular recessed portion **10***a* from the left in the axial direction, the annular recessed portion **10***a* being recessed to the right in the axial direction in a bottom portion of the recessed portion **10***b* forming the first valve chamber **13** of the valve housing **10**, and is crimped and fixed by crimping pieces **10***d* (refer to FIGS. **2** and **3**) on radially inner and outer sides formed on opening portion of the annular recessed portion **10***a*. In addition, an exposed portion of an axially left end surface of the elastic member **40** which is formed between the crimping pieces **10***d* on the radially inner and outer sides, namely, a contact surface **40***s* serves as the CS valve seat **40***a*. The contact portion **51***a* at the tip of the annular projection **51***d* of the CS valve body **51** can come into contact with and separate from the contact surface **40***s*.

In addition, the elastic member **40** and the annular recessed portion **10***a* before and after being press-fitted have a rectangular cross section, and have substantially the same dimensions on the radially inner and outer sides and in a thickness direction. Incidentally, the elastic member **40** before being press-fitted may be formed slightly larger or smaller than the annular recessed portion **10***a*.

The contact surface **40***s* of the elastic member **40** is a surface orthogonal to a driving direction of the CS valve body **51**, and a width of the contact surface **40***s* in the radial direction is formed larger than a width of the annular projection **51***d* in the radial direction. Accordingly, the contact portion **51***a* at the tip of the annular projection **51***d* can be reliably seated on the CS valve seat **40***a*, and contact between the annular projection **51***d* and the crimping pieces **10***d* on the radially inner and outer sides can be prevented.

As illustrated in FIG. **1**, the solenoid **80** mainly includes a casing **81** including an opening portion **81***a* that is open to the left in the axial direction; the center post **82** having a substantially cylindrical shape that is inserted into the opening portion **81***a* of the casing **81** from the left in the axial direction and that is fixed on a radially inner side of the casing **81**; the rod **52** which is inserted into the center post **82** to be reciprocatable in the axial direction and of which the axially left end portion is disposed on the axially left side of the CS valve seat **40***a*; the CS valve body **51** press-fitted and fixed to the axially left end portion of the rod **52**; a movable iron core **84** to which an axially right end portion of the rod **52** is inserted and fixed; a coil spring **85** as a rod biasing member (also referred to as rod biasing means) provided on an axially right side of the movable iron core **84** to bias the rod **52** inserted and fixed to the movable iron core **84**, to the left in the axial direction, namely, in the valve opening direction; and the coil **86** for excitation wound on an outer side of the center post **82** with a bobbin interposed therebetween. Incidentally, the coil spring **85** is a compression spring.

A recessed portion **81***b* of which a radially inner side of an axially left end is recessed to the right in the axial direction is formed in the casing **81**, and an axially right end portion of the valve housing **10** is inserted and fixed to the recessed portion **81***b* in a substantially sealed manner.

The center post **82** is made of a rigid body that is a magnetic material such as iron or silicon steel, and includes a cylindrical portion **82***b* which extends in the axial direction and in which an insertion hole **82***c* into which the rod **52** is inserted is formed, and the flange portion **82***d* having an annular shape and extending from an outer peripheral surface of an axially left end portion of the cylindrical portion **82***b* in the radially outward direction.

In addition, the center post **82** is inserted and fixed to the recessed portion **10***c* of the valve housing **10** inserted and fixed to the recessed portion **81***b* of the casing **81**, in a substantially sealed manner in a state where an axially right end surface of the flange portion **82***d* is brought into contact with a bottom surface of the recessed portion **81***b* of the casing **81** from the left in the axial direction. Namely, the center post **82** is fixed by sandwiching the flange portion **82***d* between the bottom surface of the recessed portion **81***b* of the casing **81** and the bottom surface of the recessed portion **10***c* of the valve housing **10** from both sides in the axial direction.

Next, an opening and closing operation of the capacity control valve V**1** will be described.

First, a non-energized state of the capacity control valve V**1** will be described. As illustrated in FIGS. **1** and **2**, in a non-energized state of the capacity control valve V**1**, the CS valve body **51** is pressed to the right in the axial direction, namely, in the valve closing direction by biasing forces of the bellows **16** and of the coil spring **17**, so that the contact portion **51***a* at the tip of the annular projection **51***d* of the CS valve body **51** is seated on the CS valve seat **40***a* formed on the axially left end surface of the elastic member **40**, and the CS valve **50** is closed.

At this time, when an effective pressure-receiving area of the bellows **16** is A, an effective pressure-receiving area of the CS valve body **51** is B, and an axially rightward direction is defined as being positive, a biasing force F_{be1} of the bellows **16**, a biasing force F_{sp1} of the coil spring **17**, a force F_{P1}) due to the control pressure Pc=(P1×(A−B)), a force F_{P2} due to the suction pressure Ps=−(P2×(A−B)), and a biasing force F_{sp2} of the coil spring **85** act on the CS valve body **51**. Namely, when the axially rightward direction is defined as being positive, a force $F_{rod}=F_{be1}+F_{sp1}+F_{P1}-F_{P2}-F_{sp2}$ acts on the CS valve body **51**.

Specifically, the fluid in the space S**1** acts on the axially left end surface of the CS valve body **51**, and the fluid in the second valve chamber **14** acts on an axially right end surface of the CS valve body **51**. Since the second valve chamber **14** and the space S**1** communicate with each other through the communication passage **51***c* formed in the CS valve body **51**, the fluid in the second valve chamber **14** located closer to a valve closing direction side than the CS valve body **51**, namely, the fluid of the suction pressure Ps supplied from the Ps port **11** flows into the space S**1**.

In addition, since the communication passage **51***c* is a narrowed through-hole, when a slight pressure difference occurs instantaneously between pressure in the space S**1** and pressure in the second valve chamber **14**, the fluid in the space S**1** is unlikely to instantaneously move toward the second valve chamber **14** and is held in the space S**1**, and a closed state of the CS valve **50** is easily maintained.

In such a manner, the fluid flowing into the space S**1** and into the second valve chamber **14** is the fluid of the same suction pressure Ps supplied from the Ps port **11**. In addition, in the present embodiment, since the effective pressure-receiving area A of the bellows **16** and the effective pressure-receiving area B of the CS valve body **51** are equal (i.e., A=B), both the forces F_{P1} and (F_{P2}) acting on the CS valve body **51** due to the control pressure Pc and to the suction pressure Ps are substantially zero. Namely, when the rightward direction is defined as being positive, a force $F_{rod}=F_{be1}+F_{sp1}-F_{sp2}$ substantially acts on the CS valve body **51**.

Next, an energized state of the capacity control valve V**1** will be described. As illustrated in FIG. **3**, in an energized state of the capacity control valve V**1**, namely, during normal control or during so-called duty control, when an electromagnetic force F_{sol} generated by the application of an electric current to the solenoid **80** is larger than the force F_{rod} (i.e., $F_{sol}>F_{rod}$), the movable iron core **84** is pulled to a center post **82** side, namely, to the axially left side, and the rod **52** fixed to the movable iron core **84** and the CS valve body **51** held so as to be able to come into contact with and to separate from the rod **52** move together to the left in the axial direction, namely, in the valve opening direction. Accordingly, the contact portion **51***a* of the CS valve body **51** separates from the CS valve seat **40***a* formed on the contact surface **40***s* of the elastic member **40**, and the CS valve **50** is opened. In addition, when the solenoid **80** is driven, the movable iron core **84** comes into contact with an axially right side of the center post **82**, so that further separation of the CS valve body **51** from the CS valve seat **40***a* is restricted.

In such a manner, the capacity control valve V**1** can control pressure in the control chamber of the variable capacity compressor using a pressure difference between the control pressure Pc and the suction pressure Ps lower than the control pressure Pc due to a valve opening degree of the CS valve **50** that is adjusted by balance between electromagnetic force of the solenoid **80** and the biasing forces of the bellows **16**, of the coil spring **17**, and of the coil spring **85**.

As described above, in the capacity control valve V**1** of the first embodiment, even in a case where contaminations are caught between the contact portion **51***a* at the tip of the annular projection **51***d* of the CS valve body **51** and the CS valve seat **40***a* formed on the contact surface **40***s* of the elastic member **40** when the valve is closed, the elastic member **40** can be elastically deformed to suppress generation of a gap between the contact portion **51***a* of the CS valve body **51** and the CS valve seat **40***a*, so that valve leakage can be reduced.

In addition, since the elastic member **40** is formed to have a modulus of elasticity, namely, a Young's modulus smaller than that of the annular projection **51***d* of the CS valve body **51**, when the valve is closed, even in a case where contaminations are caught between the contact portion **51***a* of the CS valve body **51** and the CS valve seat **40***a* formed on the contact surface **40***s* of the elastic member **40**, the elastic member **40** can be reliably and elastically deformed to suppress generation of a gap between the contact portion **51***a* of the CS valve body **51** and the CS valve seat **40***a*. Further, regardless of whether or not contaminations are caught, when the valve is closed, since the contact portion **51***a* at the tip of the annular projection **51***d* elastically deforms the elastic member **40** and is slightly buried (refer to FIG. **4**), it is difficult for the CS valve body **51** to relatively move with respect to the contact surface **40***s* of the elastic member **40** in a valve closed state of the CS valve **50**, so that sealability is enhanced.

Since the contact surface **40***s* of the elastic member **40** on which the CS valve seat **40***a* is formed is a surface orthogonal to the driving direction of the CS valve body **51**, when the valve is closed, it is difficult for the annular projection **51***d* to further relatively move with respect to the contact surface **40***s* of the elastic member **40**, so that sealability is enhanced. In addition, when the valve is closed, since the relative movement of the annular projection **51***d* with respect to the contact surface **40***s* of the elastic member **40** is suppressed, damage to the contact surface **40***s* of the elastic member **40** caused by the annular projection **51***d* is suppressed, so that sealability can be maintained over a long period of time. Incidentally, in the capacity control valve V**1** of the present embodiment, since the CS valve body **51** and the rod **52** are held so as to be able to come into contact with and to separate from each other, and the CS valve body **51** is supported by the bellows **16** fixed to the valve housing **10** via the lid member **15** on the axially left side of the CS valve seat **40***a*, the structure is such that the CS valve body **51** easily moves when the valve is closed, but as described above, the relative movement of the annular projection **51***d* with respect to the contact surface **40***s* of the elastic member **40** is suppressed, sealability or durability is guaranteed.

Further, since corners on the radially inner and outer sides of the tip portion of the annular projection **51***d* are round chamfered, and the tip portion is formed with a substantially curved cross-sectional shape, damage to the contact surface **40***s* of the elastic member **40** caused by the annular projection **51***d* is further suppressed.

When materials of the contact portion of the valve body and of the valve seat are a combination of metals as in the related art, since not only a gap is likely to be generated due to contaminations being caught, but also a gap is likely to be generated by an offset between the contact portion of the valve body and the valve seat in a closed state of the valve, valve leakage is likely to occur, but as described above, in the first embodiment, since the CS valve seat **40***a* is formed on the contact surface **40***s* of the elastic member **40**, this problem can be solved.

In addition, since the valve housing **10** is provided with the annular recessed portion **10***a*, and the elastic member **40** press-fitted to the annular recessed portion **10***a* is crimped and fixed by the crimping pieces **10***d* on the radially inner and outer sides, the elastic member **40** inserted into the annular recessed portion **10***a* can be prevented from coming off. Incidentally, since the elastic member **40** is press-fitted to the annular recessed portion **10***a* or is crimped and fixed by the crimping pieces **10***d* on the radially inner and outer sides, the elastic member **40** may be deformed in the radial direction to cause the contact surface **40***s* to slightly bulge to the left in the axial direction, so that when the valve is closed, the contact portion **51***a* at the tip of the annular projection **51***d* is easily buried in the contact surface **40***s* of the elastic member **40**.

In addition, since the elastic member **40** has a rectangular cross section, even when the annular projection **51***d* comes into contact with the contact surface **40***s* of the elastic member **40** at any position, stable elasticity can be provided.

In addition, the CS valve body **51** is configured separately from the rod **52** forming the solenoid **80**, and the rod **52** is biased in the valve opening direction by the coil spring **85**, so that the CS valve body **51** and the rod **52** are held so as to be able to come into contact with and to separate from each other. When the valve is closed, the contact portion **51***a* of the CS valve body **51** is seated on the CS valve seat **40***a* formed on the contact surface **40***s* of the elastic member **40**, and at the same time, the CS valve body **51** is not affected by an inertial force of the rod **52** acting to the right in the axial direction. For this reason, an excessive load is not applied to the annular projection **51***d* or to the elastic member **40**. Namely, when the valve is closed, since only the biasing forces of the bellows **16** and of the coil spring **17** act on the annular projection **51***d* or on the elastic member **40**, an excessive load is not applied to the annular projection **51***d* or to the elastic member **40**, so that damage thereto can be prevented.

In addition, since the coil spring **17** disposed inside the bellows **16** is a compression spring, a structure on the solenoid **80** side of the capacity control valve V**1** is simplified, and the contact portion **51***a* at the tip of the annular projection **51***d* of the CS valve body **51** is seated on the CS valve seat **40***a* while the structure is such that an axis of the CS valve body **51** is easily offset, so that good sealability can be obtained.

Incidentally, the crimping piece **10***d* may be provided on at least one of the radially inner and outer sides of the valve housing **10**.

Second Embodiment

A capacity control valve as a valve according to a second embodiment of the present invention will be described with reference to FIG. **5**. Incidentally, a description of configurations that are the same as the configurations of and are duplicated in the first embodiment will be omitted.

As illustrated in FIG. **5**, in a capacity control valve V**2** in the second embodiment, a valve housing is configured such that a first valve housing **210** and a second valve housing **211** are integrally connected and fixed to each other in a substantially sealed state by externally fitting an axially left end portion of the second valve housing **211** to an axially right end portion of the first valve housing **210** from the right in the axial direction. The Pc port **12** communicating with the control chamber of the variable capacity compressor is formed in the first valve housing **210**. In addition, the Ps port **11** communicating with the suction chamber of the variable capacity compressor is formed in the second valve housing **211**.

An annular recessed portion **211***a* recessed to the right in the axial direction is formed in the axially left end portion of the second valve housing **211**, and an elastic member **240** is press-fitted to the annular recessed portion **211***a* from the left in the axial direction, and is crimped and fixed by a crimping piece **211***d* on a radially inner side formed on an opening portion of the annular recessed portion **211***a*. Further, when the first valve housing **210** and the second valve housing **211** are connected and fixed to each other, an annular protrusion **210***a* having a rectangular cross section that is formed on the axially right end portion of the first valve housing **210** is pressed against a radially outer side of an axially left end surface of the elastic member **240**, so that the elastic member **240** is sandwiched and held between the first valve housing **210** and the second valve housing **211**.

In addition, a CS valve seat **240***a* as a valve seat is formed on the axially left end surface of the elastic member **240**, the CS valve seat **240***a* being formed of an exposed portion of the axially left end surface formed between the annular protrusion **210***a* of the first valve housing **210** and the crimping piece **211***d* of the second valve housing **211**, namely, of a contact surface **240***s*.

According to this configuration, in the capacity control valve V**2** of the second embodiment, since the elastic member **240** is sandwiched and held between the first valve housing **210** and the second valve housing **211** that are integrally connected and fixed to each other, the elastic member **240** inserted into the annular recessed portion **211***a* can be prevented from coming off.

In addition, compared to when the crimping pieces **10***d* are formed on the radially inner and outer sides inside the valve housing **10** as in the first embodiment, in the second embodiment, the valve housing is divided, so that processing of the first valve housing **210** and the second valve housing **211** for holding the elastic member **240** is facilitated.

Incidentally, in the second embodiment, since the elastic member **240** can be held by the annular protrusion **210***a* of the first valve housing **210**, the crimping piece **211***d* of the second valve housing **211** may not be formed. Accordingly, only a recess of the annular recessed portion **211***a* may be formed in the second valve housing **211**, and the second valve housing **211** is more easily processed.

Third Embodiment

A capacity control valve as a valve according to a third embodiment of the present invention will be described with reference to FIGS. **6** to **8**. Incidentally, a description of configurations that are the same as the configurations of and are duplicated in the first embodiment will be omitted.

As illustrated in FIG. **6**, in a capacity control valve V**3** of the third embodiment, the CS valve **50** is formed of a CS valve body **351** as a valve body and a CS valve seat **310***a* as a valve seat at a tip of an annular projection **310***d* of a valve housing **310**, and a contact portion **340***a* formed on a contact surface **340***s* of an elastic member **340** that is press-fitted and crimped and fixed to an annular recessed portion **351***a* formed at an axially right end of the CS valve body **351** comes into contact with and separates from the CS valve seat **310***a* in the axial direction, to open and close the CS valve **50**.

As illustrated in FIGS. **6** to **8**, the CS valve body **351** is made of a metallic material, and the annular recessed portion **351***a* recessed to the left in the axial direction is formed in the CS valve body **351** at a position that is much further offset in the radially outward direction than a communication passage **351***c*.

The elastic member **340** is press-fitted to the annular recessed portion **351***a* of the CS valve body **351** from the right in the axial direction, and is crimped and fixed by crimping pieces **351***d* (refer to FIGS. **7** and **8**) on radially inner and outer sides formed on an opening portion of the annular recessed portion **351***a*. In addition, the contact portion **340***a* of the CS valve body **351** is formed on an axially right end surface of the elastic member **340**, the contact portion **340***a* being formed of an exposed portion of the axially right end surface formed between the crimping pieces **351***d* on the radially inner and outer sides, namely, of the contact surface **340***s*. The contact portion **340***a* can come into contact with and separate from the CS valve seat **310***a* at the tip of the annular projection **310***d* of the valve housing **310**.

As illustrated in FIGS. **6** to **8**, the annular projection **310***d* that is formed on a bottom portion of a recessed portion **310***b* forming the first valve chamber **13** and that protrudes to the left in the axial direction is formed on the valve housing **310**. The CS valve seat **310***a* from which the contact portion **340***a* of the CS valve body **351** comes into contact with and separates in the axial direction is formed at the tip, namely, an axially left end of the annular projection **310***d*.

According to this configuration, in the capacity control valve V**3** of the third embodiment, since the elastic member **340** is provided on the CS valve body **351** that is a member to be assembled to the valve housing **310**, processing of the CS valve body **351** for holding the elastic member **340** is facilitated.

Incidentally, the crimping piece **351***d* may be provided on at least one of the radially inner and outer sides of the CS valve body **351**.

The embodiments of the present invention have been described above with reference to the drawings, but the specific configurations are not limited to the embodiments, and changes or additions that are made without departing from the scope of the present invention are also included in the present invention.

For example, in the embodiments, a mode in which one of the CS valve seat and the contact portion of the CS valve body facing the annular projection in the axial direction is formed of an elastic member has been described, but the present invention is not limited to the configuration, and both the contact portion of the CS valve body and the CS valve seat may be formed of elastic members. In this case, only the annular projection may be formed of an elastic member. In addition, it is preferable that the elastic member on which the contact surface is formed has a modulus of elasticity smaller than that of the elastic member forming the annular projection. Incidentally, both the contact portion of the CS valve body and the CS valve seat may be formed of elastic members having the same modulus of elasticity.

In addition, the contact surface of the elastic member may not be a surface orthogonal to the driving direction of the CS valve body, and may be formed as, for example, a tilted surface or a curved surface.

In addition, in the embodiments, the valve housing and the CS valve body have been described as being made of a metallic material or a resin material, but it is preferable that a member by which the elastic member is held through crimping and fixing is made of a metallic material. In addition, for example, when a member by which the elastic member is held is made of a resin material, instead of the crimping piece, a portion that becomes a pressing piece that prevents the elastic member from coming off may be fleshed out.

In addition, in the embodiments, the elastic member has been described as being press-fitted to the annular recessed portion, but the elastic member may be simply inserted into the annular recessed portion. In addition, the elastic member is not limited to being disposed inside the annular recessed portion, and for example, may be fixed to the valve housing or to the CS valve body by another member such as an adhesive agent or a bolt.

In addition, the elastic member is not limited to having a rectangular cross section, and the cross-sectional shape may be, for example, a circular shape, a triangular shape, a T shape, a V shape, an X shape, etc. For example, as illustrated in FIG. **9**, when an elastic member **40'** is formed with a T-shaped cross section, the elastic member **40'** may be crimped and fixed by pressing step portions provided in the elastic member **40'** press-fitted to the annular recessed portion **10***a*, using the crimping pieces **10***d* on the radially inner and outer side.

In addition, in the embodiments, the CS valve body has been described as being configured as a member separate from the rod disposed through the coil of the solenoid, and as being able to come into contact with and to separate from the rod, but the present invention is not limited to the configuration, and the CS valve body and the rod that are separate members may be integrally connected and fixed to each other. In addition, the CS valve body and the rod may be integrally formed. In this case, the coil spring that biases the rod in the valve opening direction may not be provided.

In addition, since the coil spring is disposed inside the bellows, the bellows itself may not have a biasing force.

In addition, the coil spring may not be disposed inside the bellows.

In addition, in the embodiments, a mode in which the effective pressure-receiving area A of the bellows and the effective pressure-receiving area B of the CS valve body are the same (i.e., A=B) has been described, but the present invention is not limited to this mode, and the effective pressure-receiving area A may be set slightly larger than the effective pressure-receiving area B (i.e., A>B) such that a closed state of the CS valve can be reliably maintained, or the effective pressure-receiving area B may be set slightly larger than the effective pressure-receiving area A (i.e., A<B) such that the CS valve is easily opened. Namely, an influence of pressure of the fluid acting on both sides the CS valve body in a movement direction may be reduced.

In addition, the capacity control valve of the embodiments has been described as a CS valve, but the present invention is not limited to the configuration, and the capacity control valve may be a DC valve that opens and closes a flow passage between the Pd port and the Pc port.

In addition, the drive source may be a member other than the solenoid.

REFERENCE SIGNS LIST

9 Fixed orifice
10 Valve housing
10*a* Annular recessed portion
10*d* Crimping piece
11 Ps port (port)
12 Pc port (port)
13 First valve chamber
14 Second valve chamber
15 Lid member
16 Bellows (biasing member)
17 Coil spring (biasing member)
40 Elastic member
40*a* CS valve seat (valve seat)
40*s* Contact surface
50 CS valve
51 CS valve body (valve body)
51*a* Contact portion
51*d* Annular projection
52 Rod
80 Solenoid (drive source)
85 Coil spring (rod biasing member)
210 First valve housing (valve housing)
210*a* Annular protrusion
211 Second valve housing (valve housing)
211*a* Annular recessed portion
211*d* Crimping piece
240 Elastic member
240*a* CS valve seat (valve seat)
240*s* Contact surface
310 Valve housing
310*a* CS valve seat (valve seat)
310*d* Annular projection
340 Elastic member
340*a* Contact portion
340*s* Contact surface
351 CS valve body (valve body)
351*a* Annular recessed portion
351*d* Crimping piece
S1 Space
V**1**, V**2**, V**3** Capacity control valve (valve)

The invention claimed is:

1. A valve, comprising:
a valve housing in which a port through which a fluid passes is formed;
a valve body configured to be driven by a drive source;
a valve seat on which a contact portion of the valve body is seated;
a biasing member configured to bias the valve body in a valve closing direction of the valve; and
a bellows having a first end portion fixed to the valve body and a second end portion fixed to the valve housing, the first end portion and the second end portion being opposed to each other in an axial direction,
wherein the valve seat is formed of an elastic member which is press-fitted into an annular recessed portion formed in the valve housing,
the annular recessed portion is defined by an annular bottom surface, and an inner annular side wall surface and an outer annular side wall surface which are opposed to each other in a radial direction,
the contact portion of the valve body is formed of an annular projection which is protruded toward the valve seat from a central part of an axially end surface of the valve body on a valve seat side and which has a contact surface formed by a flat tip surface orthogonal to the axial direction, and
the valve body is provided with a through-hole which communicates with a radially inner side space of the contact portion and an inner space of the bellows.

2. The valve according to claim **1**,
the elastic member is formed to have a modulus of elasticity smaller than a modulus of elasticity of the annular projection.

3. The valve according to claim **2**,
wherein an inner diameter side end of the elastic member is crimped and fixed.

4. The valve according to claim **2**,
wherein the elastic member has a rectangular cross section.

5. The valve according to claim **2**,
wherein the valve body is separately configured to come into contact with and to separate from a rod forming the drive source, and
the rod is biased in a valve opening direction by a rod biasing member.

6. The valve according to claim **2**,
wherein the biasing member is a compression spring.

7. The valve according to claim **1**,
wherein an inner diameter side end of the elastic member is crimped and fixed.

8. The valve according to claim **7**,
wherein the elastic member has a rectangular cross section.

9. The valve according to-claim **7**,
wherein the valve body is separately configured to come into contact with and to separate from a rod forming the drive source, and
the rod is biased in a valve opening direction by a rod biasing member.

10. The valve according to claim **7**,

wherein the biasing member is a compression spring.

11. The valve according to claim **1**,

wherein the elastic member has a rectangular cross section.

12. The valve according to claim **11**,

wherein the valve body is separately configured to come into contact with and to separate from a rod forming the drive source, and

the rod is biased in a valve opening direction by a rod biasing member.

13. The valve according to claim **11**,

wherein the biasing member is a compression spring.

14. The valve according to claim **1**,

wherein the valve body is separately configured to come into contact with and to separate from a rod forming the drive source, and

the rod is biased in a valve opening direction by a rod biasing member.

15. The valve according to claim **1**,

wherein the biasing member is a compression spring.

* * * * *